ECHOBRIDGE: DUAL-ENCODER AUDIO FUSION BASED ON RESIDUAL TIME ADAPTATION

Technical Report

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ABSTRACT

This report presents Echobridge, a dual-encoder model submitted to the 2026 Interspeech Audio Encoder Capacity Challenge. The challenge evaluates audio-encoder performance across datasets covering speech, environmental sounds, and music. Recognising that single-encoder models often specialize in specific audio domains, we integrate representations from Whisper (speech) and CED-mini (environmental sounds) using a zero-initialised multi-scale temporal convolutional adapter, which enables adaptive fusion via channel-wise weighting. Experiments across multiple datasets demonstrate that Echobridge outperforms single-encoder baselines on classification tasks. In particular, it achieves performance gains of 11.4% over CED-mini and 6.2% over Whisper on the audio-classification task.

***Index Terms*—** Echobridge, dual-encoder, multi-scale temporal convolutional adapter, audio-classification task

1. INTRODUCTION

Most existing encoders are optimized for a single dominant audio domain, which results in poor generalisation capabilities when applied to heterogeneous audio scenarios containing ambient sounds, music, and non-speech acoustic events. To systematically investigate this limitation, the 2026 Interspeech Audio Encoder Capacity Challenge (AECC) focuses on evaluating the ability of audio encoders to generalise across different datasets and tasks [1]. The Challenge emphasises encoder-centric evaluation, i.e. applying the same downstream classifiers to different fixed or lightly tuned encoders.

Speech-oriented encoders, such as Whisper [2], excel at modelling linguistic and phonological structures, but are less sensitive to non-speech acoustic cues. In contrast, encoders trained on audio annotations or ambient acoustic datasets, such as CED-based models [3], are more effective at capturing background texture and acoustic events, but lack fine-grained temporal modelling of speech. This inherent specialisation suggests that no single encoder can be universally optimal in all audio domains.

In this report, we propose a dual-encoder audio representation model termed as Echobridge, which is designed for the AECC testing environment. Echobridge uses a strict linear splicing fusion as a baseline to preserve the original representation capabilities of both encoders. On top of this baseline, we introduce a zero-initialisation multi-scale temporal convolution adapter as a lightweight residual module. This design ensures that the initial behaviour of the model is identical to the baseline fusion, while allowing for adaptive optimisation through channel-level modulation and multiscale temporal context modelling during training.

Extensive experiments on the AECC benchmark dataset show that Echobridge consistently outperforms single-encoder baseline methods on audio classification tasks. Notably, our approach achieves significant improvements over both Whisper and CED-mini, highlighting the effectiveness of complementary encoder fusion and carefully constrained adaptation.

2. SYSTEM DESCRIPTIONS

2.1. Background

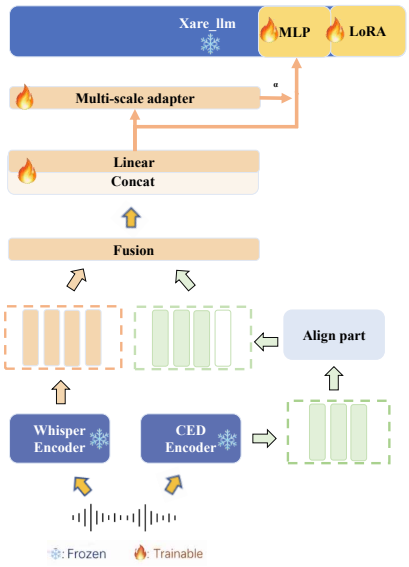


Figure 1: Echobridge Architecture.

In this Challenge, we selected Whisper [2] and CED-mini [3] as complementary encoders because they exhibit different inductive

biases and are trained with distinct objectives. Whisper is trained on large-scale speech data and excels at modelling linguistic content and speech structure, making it highly effective for speech-oriented tasks. By contrast, CED-mini is designed for general-purpose acoustic event modelling and captures rich environmental and other non-speech acoustic information. By leveraging these two encoders, we aim to integrate speech-centric and environment-centric acoustic knowledge within a unified framework.

As shown in Figure 1, raw audio waveforms are processed in parallel by a frozen Whisper encoder and a frozen CED-mini encoder. Because the two encoders operate at different temporal resolutions, we first align their frame-level outputs along the temporal dimension via interpolation, using the Whisper output as the reference time scale. The aligned features are then projected into a shared embedding space and fused through splicing followed by a linear transformation, which serves as a robust and stable baseline fusion strategy.

To enhance the fused representation while preserving the baseline behaviour, we introduce a lightweight multi-scale temporal adapter on top of the fused features. The adapter comprises depthwise separable temporal convolutions with multiple kernel sizes, enabling it to capture complementary short- and long-range temporal patterns. More importantly, the adapter output is scaled by a learnable, channel-wise coefficient initialised to zero, ensuring that the model starts from a strictly linear splicing baseline and departs from it only when beneficial during training [4].

During downstream evaluations, both encoders remain frozen, and only the fusion layer and adapters are trained [5]. This design enables adaptive optimisation of the fused representation with minimal additional parameters, while maintaining strong generalisation across a range of AECC tasks.

3. METHODS

3.1. Dual-Encoder Front End

For our Challenge submission, we combine two complementary encoders. We employ Whisper for its strong performance on the X-ARES benchmark in speech-focused evaluations and pair it with CED, which has demonstrated excellent general-domain capability across a broad range of acoustic conditions [6].

Given an input waveform $X \in R^{B\times L}$, where B is the batch size and L is the number of audio samples, two encoders produce frame-level representations, given by

$$H^{(w)} = f_{whisper}(X) \in R^{B\times T_w\times D_w}, \tag{1}$$

$$H^{(c)} = f_{CED}(X) \in R^{B\times T_c\times D_c}, \tag{2}$$

where T_w and T_c denote the temporal resolutions of Whisper and CED-mini, respectively, D_w and D_c are the feature dimensions.

3.2. Temporal Alignment

As Whisper and CED-mini operate at different temporal resolutions, their frame-level representations are not inherently aligned along the time axis. In particular, Whisper produces representations at a relatively coarse temporal stride due to its convolutional front-end, whereas CED-mini yields denser temporal outputs, as dictated by its patch-based spectro-temporal encoding [7]. To facilitate effective fusion, we therefore perform explicit temporal alignment of the two feature sequences.

The frame-level representations produced by two encoders are given in (1) and (2). We use the Whisper representation as the reference time scale, since it more directly reflects linguistic content and the temporal structure of speech. When $T_c \neq T_w$, we resample the CED represesntation along the temporal dimension to match T_w [8].

Concretely, we apply linear interpolation to the CED features:

$$\widetilde{H}^{(c)} = Interp(H^c, T_w), \tag{3}$$

where $Interp(\cdot)$ denotes 1D interpolation along the temporal axis and the resulting tensor is $\widetilde{H}^{(c)} \in R^{B\times T_w\times D_w}$.

This operation preserves the global temporal structure of acoustic events while ensuring compatibility with the Whisper time grid. The corresponding attention masks are aligned via nearest-neighbour interpolation to preserve valid frame boundaries.

After alignment, two encoders yield synchronised frame-level representations, $H^{(w)}$ and $\widetilde{H}^{(c)}$, with an identical temporal length of T_w, thereby enabling subsequent fusion at each time step. This design allows the model to integrate speech-centric and environment-centric cues in a temporally coherent manner, without modifying the internal architectures of either pretrained encoder [5].

3.3. Residual Multi-scale Temporal Adapter

After baseline fusion, we introduce a lightweight residual temporal adapter to refine the fused representation, while strictly preserving the original fusion behaviour at initialisation [4]. Given the fused sequence representation $H^{(f)} \in R^{B\times T\times D}$ obtained via concat-linear fusion, the adapter predicts a residual update that is added back to the base features.

To that end, we employ multiple depthwise temporal convolution operators with different kernel sizes $k \in \mathcal{K}$, where $\mathcal{K} = \{k_1, k_2\}$ and $k_1 < k_2$ (e.g, $k_1 = 3, k_2 = 7$), and each convolution operates independently on each channel [9]. That is,

$$\Delta_k = DWConv_k(LN(H), k \in \mathcal{K}, \tag{4}$$

where $DWConv_k(\cdot)$ denotes a depthwise 1D convolution along the temporal dimension with kernel size k.

This design allows the adapter to capture both short-range and longer-range temporal patterns without introducing cross-channel interference. The multi-scale responses are then aggregated and projected back to the original feature dimension using a pointwise convolution as

$$\Delta = PWConv(\sum_{k\in\mathcal{K}} \Delta_k), \tag{5}$$

where $PWConv(\cdot)$ is a 1×1 convolution applied across channels. It is of interest to point out that the weights of this projection layer are initialised to zero, ensuring that $\Delta = 0$ at initialisation.

To further control how this residual update contributes across channels, we employ a squeeze–excitation style gating mechanism. A global temporal average pooling is applied to summarise each channel's activation over time, followed by a lightweight gating network, given by

$$g = \tanh(W_2\sigma(W_1 AvgPool_t(H))), \tag{6}$$

where $g \in \mathcal{D}$ is a channel-wise modulation vector. This gate allows the model to adaptively emphasise or suppress the residual correction on a per-channel basis, depending on the global temporal characteristics of the fused sequence. The final projection layer of the

Table 1: X-ARES Track A local evaluation results, where bold and underlined results are the first and second-place scores.

Task	Baseline (base) Dasheng	CED-mini	Whisper	+Fusion	+concatlinear	+Adapter
Speech tasks						
speech commands	<u>0.655</u>	0.258	**0.703**	0.599	0.606	0.632
LibriCount	0.386	0.406	0.331	0.380	<u>0.440</u>	**0.470**
VoxLingua33	0.311	0.200	**0.848**	<u>0.797</u>	0.778	0.779
VoxCeleb1-Binary	**0.974**	<u>0.934</u>	0.767	0.878	0.874	0.886
ASVspoof2015	**0.937**	0.819	<u>0.935</u>	0.896	0.899	0.909
Fluent Speech Commands	**0.984**	0.245	<u>0.818</u>	0.781	0.773	0.771
VocalSound	0.855	0.894	0.865	0.910	**0.914**	<u>0.913</u>
CREMA-D	**0.621**	0.490	<u>0.528</u>	0.499	0.494	0.490
Environmental sound tasks						
ESC-50	0.755	0.853	0.672	<u>0.875</u>	0.858	**0.877**
FSD50k	0.063	**0.658**	0.094	0.200	<u>0.205</u>	0.198
UrbanSound 8k	0.829	0.841	0.724	<u>0.852</u>	**0.882**	0.830
FSD18-Kaggle	0.415	<u>0.624</u>	0.530	**0.683**	**0.683**	**0.683**
Music tasks						
GTZAN Genre	0.323	<u>0.889</u>	0.737	0.838	0.879	**0.899**
NSynth-Instruments	0.675	0.674	0.606	**0.705**	<u>0.695</u>	0.693
Free Music Archive Small	0.429	0.629	0.595	0.632	<u>0.642</u>	**0.646**
Unweighted Average	0.614	0.598	0.650	0.702	<u>0.708</u>	**0.712**

Table 2: X-ARES Track B local evaluation results, where bold and underlined results are the first and second-place scores.

Task	Baseline (base) Dasheng	CED-mini	Whisper	+Fusion	+concatlinear	+Adapter
Understanding tasks						
LibriSpeech-100h	0.000	0.000	**0.354**	0.270	<u>0.319</u>	0.302
AISHELL-1-100h	0.000	0.000	**0.304**	0.177	0.208	<u>0.236</u>
Clotho	0.220	<u>0.433</u>	0.332	**0.440**	0.429	0.422
The Song Describer Dataset	0.399	<u>0.469</u>	0.457	0.462	0.467	**0.477**
MECAT	0.529	0.529	0.548	**0.572**	0.567	<u>0.569</u>
Unweighted Average	0.229	0.286	<u>0.399</u>	0.384	0.398	**0.401**

gating network is zero-initialised, ensuring that $g = 0$ at initialisation.

The gated residual update is then incorporated via a residual connection as

$$H' = H + \alpha \odot (\Delta H \odot (1 + g)), \tag{7}$$

where $\alpha \in R^D$ is a learnable channel-wise scaling parameter initialised to zero. This design guarantees that the entire adapter path is functionally inactive at initialisation, making the model strictly equivalent to the concat–linear baseline at the start of training.

As training progresses, the residual adapter gradually learns to introduce temporally structured and channel-selective refinements. This allows the model to adaptively enhance the fused representation while maintaining the stability and strong performance of the baseline fusion strategy [10].

4. RESULTS AND DISCUSSIONS

Across the evaluation benchmarks of the X-ARES Audio Encoder Capacity Challenge, our proposed dual-encoder model equipped with a multi-scale temporal convolutional adapter demonstrates strong and consistent performance on both Track A and Track B.

For Track A, which focuses on audio classification across the domains of speech, environmental sound, and music, our final submission (+Adapter) achieves the highest unweighted average score of 0.712, outperforming the Challenge baseline Dasheng (0.614) as well as strong single-encoder baselines, Whisper (0.650) and CED-mini (0.598). As summarised in Table 1, the proposed model effectively integrates the complementary strengths of two encoders, yielding robust improvements across diverse tasks. In particular, it maintains Whisper's advantage on speech-centric tasks such as VoxLingua33 and ASVspoof2015, while preserving CED-mini's strengths on environmental-sound and music datasets, including ESC-50 and GTZAN Genre. Moreover, the consistent gains from +Fusion to +Concat+Linear, and finally to +Adapter, further support the effectiveness of the proposed adapter in refining the fused representation.

For Track B, which evaluates higher-level audio-understanding tasks requiring semantic and cross-modal reasoning, the proposed model again attains the best overall performance, achieving an unweighted average score of 0.401, shown in Table 2. Compared with Whisper (0.399) and the concat-linear fusion baseline (0.398), the +Adapter model delivers consistent improvements across most datasets, including AISHELL-1-100h and the Song De-

scriber Dataset. These results indicate that the refined temporal representations learned by the adapter not only benefit low-level acoustic classification, but also transfer effectively to more abstract, understanding-oriented tasks.

In a conclusion, results on both tracks demonstrate that the proposed dual-encoder architecture, augmented with a zero-initialised multi-scale temporal adapter, provides a strong and general-purpose audio representation. It achieves balanced performance across heterogeneous audio domains and task types, supporting its suitability as a robust audio front-end for large-scale audio-language modelling.

5. CONCLUSION

In this report, we propose Echobridge to address the need for a general-purpose and robust audio front-end for large-scale audio language models. Echobridge combines the Whisper encoder, which specialises in speech content, with the CED-mini encoder, which is effective for modelling ambient sounds, and further introduces a zero-initialised multi-scale temporal convolutional adapter to enable deep feature fusion while ensuring stable optimisation. This design yields a robust dual-encoder architecture that integrates complementary inductive biases from heterogeneous pretrained encoders. Under the standardised evaluation protocol of the Interspeech 2026 Audio Encoder Capacity Challenge, Echobridge achieves strong and well-balanced performance across a broad set of classification tasks spanning speech, ambient sound, and music, attaining a composite score that exceeds both the challenge baseline and strong single-encoder references. These results support the effectiveness of fusing heterogeneous pretrained models to leverage complementary strengths, and highlight the importance of the proposed adapter in improving cross-domain audio semantic representation.

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